Survivor Analytics

Jaden Rothschild, Rowan Rosenblum

2026-05-05

```
# ============================================================
# SURVIVOR NONPARAMETRIC ANALYSIS - ALL 7 RESEARCH QUESTIONS
# ============================================================
# Required packages: none beyond base R
# setwd() to the folder containing all 6 CSV files before running
#
# Plots display on screen as the script runs.
# ============================================================

# ---- shared plot helper colors ----
col_ret  <- "#E07B54"   # returnee / group 1
col_new  <- "#5B8DB8"   # new / group 2
col_mid  <- "#888888"   # reference line
col_bg   <- "#F7F7F7"
par(ask = TRUE)   # press Enter in console to advance to next plot


# ============================================================
# QUESTION 1: Does original season placement affect how you
#             place on an all-returning season?
#             Seasons as blocks: 8, 20, 31, 34, 50
#             Test: Spearman Rank Correlation
# ============================================================
cat("\n", strrep("=", 60), "\n")
```

```
##
##  ============================================================
```

```
cat("QUESTION 1: Spearman Rank Correlation\n")
```

```
## QUESTION 1: Spearman Rank Correlation
```

```
cat("Does past placement predict all-star placement?\n")
```

```
## Does past placement predict all-star placement?
```

```
cat(strrep("=", 60), "\n\n")
```

```
## ============================================================
```

```r
q1 <- read.csv("Survivor_Q1.csv")

cat("--- Assumption 1: Normality (Shapiro-Wilk) ---\n")
```

```
## --- Assumption 1: Normality (Shapiro-Wilk) ---
```

```r
cat("H0: Data is normally distributed\n\n")
```

```
## H0: Data is normally distributed
```

```r
sw_placement <- shapiro.test(q1$Placement_CDF)
sw_past      <- shapiro.test(q1$Past_Placement_CDF)
cat(sprintf("Placement CDF:      W = %.4f, p = %.4f\n",
            sw_placement$statistic, sw_placement$p.value))
```

```
## Placement CDF:      W = 0.9461, p = 0.0004
```

```r
cat(sprintf("Past Placement CDF: W = %.4f, p = %.4f\n\n",
            sw_past$statistic, sw_past$p.value))
```

```
## Past Placement CDF: W = 0.9165, p = 0.0000
```

```r
if (sw_placement$p.value < 0.05 | sw_past$p.value < 0.05) {
  cat("CONCLUSION: At least one variable violates normality (p < 0.05).\n")
  cat("Pearson correlation is NOT appropriate.\n")
  cat("=> Nonparametric Spearman Rank Correlation is required.\n\n")
} else {
  cat("Both variables appear normal; Spearman is appropriate given\n")
  cat("the ordinal nature of placement data.\n\n")
}
```

```
## CONCLUSION: At least one variable violates normality (p < 0.05).
## Pearson correlation is NOT appropriate.
## => Nonparametric Spearman Rank Correlation is required.
```

```r
cat("--- Overall Spearman Rank Correlation (all seasons pooled) ---\n")
```

```
## --- Overall Spearman Rank Correlation (all seasons pooled) ---
```

```r
overall <- cor.test(q1$Placement_CDF, q1$Past_Placement_CDF,
                    method = "spearman", exact = FALSE)
cat(sprintf("rho = %.4f,  S = %.1f,  p = %.4f\n\n",
            overall$estimate, overall$statistic, overall$p.value))
```

```
## rho = -0.2127,  S = 214468.2,  p = 0.0318
```

```r
cat("--- Spearman Correlation by Season (Block) ---\n")
```

```
## --- Spearman Correlation by Season (Block) ---
```

```r
cat(sprintf("%-10s %-5s %-8s %-8s\n", "Season", "n", "rho", "p-value"))
```

```
## Season     n     rho      p-value
```

```r
cat(strrep("-", 35), "\n")
```

```
## -----------------------------------
```

```r
season_rhos   <- numeric(0)
season_labels <- character(0)
for (s in sort(unique(q1$Season))) {
  sub <- q1[q1$Season == s, ]
  r   <- cor.test(sub$Placement_CDF, sub$Past_Placement_CDF,
                  method = "spearman", exact = FALSE)
  cat(sprintf("%-10s %-5d %-8.4f %-8.4f\n",
              paste0("S", s), nrow(sub), r$estimate, r$p.value))
  season_rhos   <- c(season_rhos, r$estimate)
  season_labels <- c(season_labels, paste0("S", s))
}
```

```
## S8         18    -0.7618  0.0002
## S20        20    0.1879   0.4275
## S31        20    -0.0799  0.7377
## S34        20    -0.2831  0.2264
## S50        24    -0.2652  0.2105
```

```r
cat("\nINTERPRETATION:\n")
```

```
##
## INTERPRETATION:
```

```r
cat("- Overall rho is negative and significant: players who placed\n")
```

```
## - Overall rho is negative and significant: players who placed
```

```r
cat("  better in the past do NOT reliably place better in all-stars.\n")
```

```
##   better in the past do NOT reliably place better in all-stars.
```

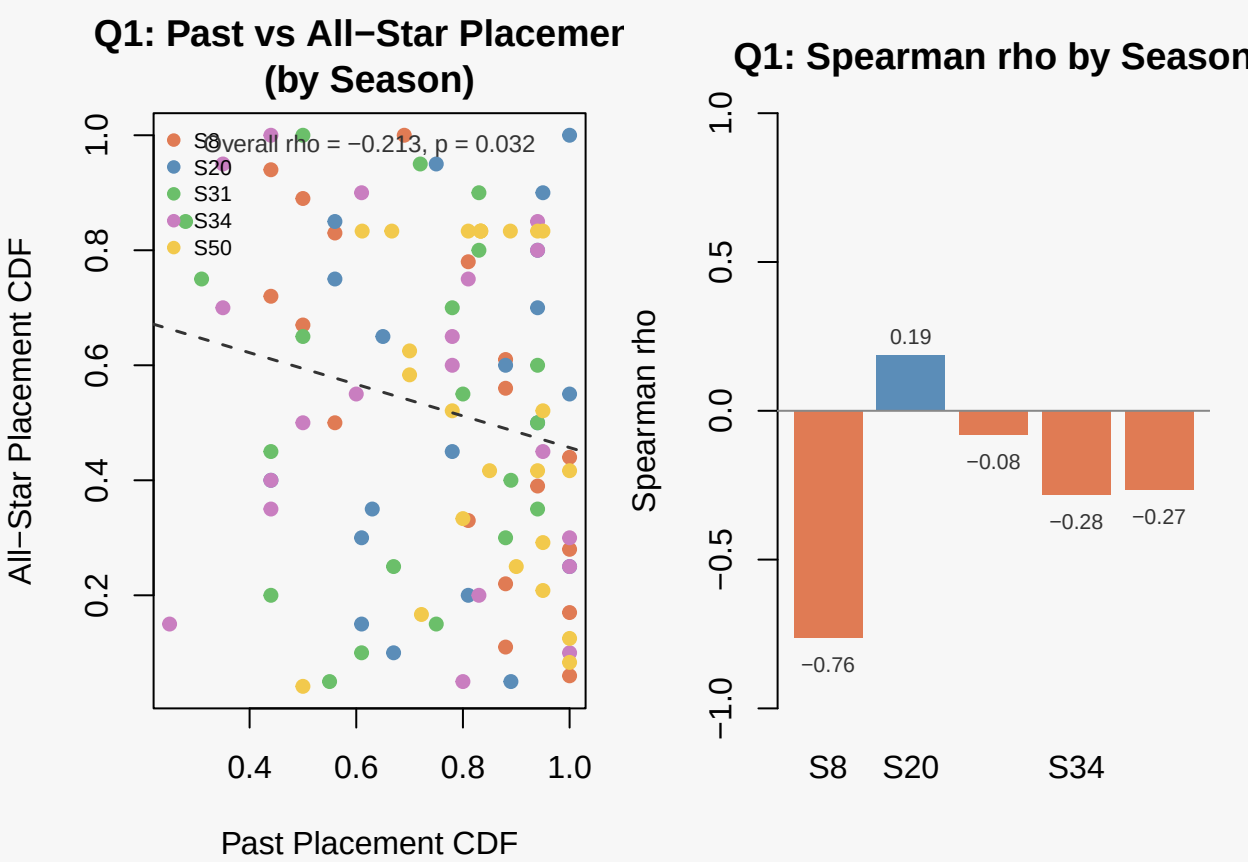
```r
cat("- Season 8 shows a strong significant negative correlation.\n")
```

```
## - Season 8 shows a strong significant negative correlation.
```

```r
cat("- Remaining seasons show weak, non-significant correlations.\n")
```

```
## - Remaining seasons show weak, non-significant correlations.
```

```r
# --- Q1 PLOT ---
par(mfrow = c(1, 2), bg = col_bg, mar = c(4, 4, 3, 1))
season_ids <- sort(unique(q1$Season))
pcols <- colorRampPalette(c("#E07B54","#5B8DB8","#6BBF6A","#C97EC0","#F2C94C"))(length(season_ids))
plot(q1$Past_Placement_CDF, q1$Placement_CDF,
     col = pcols[match(q1$Season, season_ids)],
     pch = 19, cex = 0.9,
     xlab = "Past Placement CDF", ylab = "All-Star Placement CDF",
     main = "Q1: Past vs All-Star Placement\n(by Season)")
abline(lm(Placement_CDF ~ Past_Placement_CDF, data = q1),
       col = "#333333", lty = 2, lwd = 1.5)
legend("topleft", legend = paste0("S", season_ids),
       col = pcols, pch = 19, pt.cex = 0.8, cex = 0.7, bty = "n")
mtext(sprintf("Overall rho = %.3f, p = %.3f",
              overall$estimate, overall$p.value),
      side = 3, line = -1.2, cex = 0.75, col = "#333333")
bar_cols <- ifelse(season_rhos < 0, col_ret, col_new)
bp <- barplot(season_rhos, names.arg = season_labels,
              col = bar_cols, border = NA, ylim = c(-1, 1),
              ylab = "Spearman rho",
              main = "Q1: Spearman rho by Season")
abline(h = 0, col = col_mid, lwd = 1)
text(bp, season_rhos + ifelse(season_rhos >= 0, 0.06, -0.09),
     round(season_rhos, 2), cex = 0.7, col = "#333333")
```




```
# ============================================================
# QUESTION 2: How do past winners perform in a new season?
#             n = 11, Wilcoxon Signed-Rank + Sign Test
# ============================================================
cat("\n\n", strrep("=", 60), "\n")
```

```
##
##
##  ============================================================
```

```
cat("QUESTION 2: Wilcoxon Signed-Rank Test & Sign Test\n")
```

```
## QUESTION 2: Wilcoxon Signed-Rank Test & Sign Test
```

```
cat("Do past winners perform differently than average (CDF = 0.5)?\n")
```

```
## Do past winners perform differently than average (CDF = 0.5)?
```

```
cat(strrep("=", 60), "\n\n")
```

```
## ============================================================
```

```r
q2 <- read.csv("Survivor_Q2.csv")

cat("--- Assumption 1: Normality (Shapiro-Wilk) ---\n")
```

```
## --- Assumption 1: Normality (Shapiro-Wilk) ---
```

```r
sw_q2 <- shapiro.test(q2$Placement_CDF)
cat(sprintf("Placement CDF: W = %.4f, p = %.4f\n\n",
            sw_q2$statistic, sw_q2$p.value))
```

```
## Placement CDF: W = 0.9183, p = 0.3050
```

```r
if (sw_q2$p.value < 0.05) {
  cat("CONCLUSION: Placement CDF is NOT normally distributed (p < 0.05).\n")
  cat("=> Nonparametric tests required.\n\n")
} else {
  cat("NOTE: p > 0.05 but n = 11 is small; Shapiro-Wilk has low power.\n")
  cat("With small n, nonparametric tests are more appropriate.\n\n")
}
```

```
## NOTE: p > 0.05 but n = 11 is small; Shapiro-Wilk has low power.
## With small n, nonparametric tests are more appropriate.
```

```r
cat("--- Assumption 2: Sample Size ---\n")
```

```
## --- Assumption 2: Sample Size ---
```

```r
cat(sprintf("n = %d  (small sample; nonparametric tests preferred)\n\n", nrow(q2)))
```

```
## n = 11  (small sample; nonparametric tests preferred)
```

```r
mu0   <- 0.5
diffs <- q2$Placement_CDF - mu0

cat("--- Wilcoxon Signed-Rank Test ---\n")
```

```
## --- Wilcoxon Signed-Rank Test ---
```

```r
cat("H0: Median placement CDF = 0.5 (winners perform at average)\n")
```

```
## H0: Median placement CDF = 0.5 (winners perform at average)
```

```r
cat("H1: Median placement CDF != 0.5\n\n")
```

```
## H1: Median placement CDF != 0.5
```

```r
wsr <- wilcox.test(q2$Placement_CDF, mu = mu0, exact = FALSE)
cat(sprintf("V = %.1f,  p = %.4f\n\n", wsr$statistic, wsr$p.value))
```

```
## V = 26.0,  p = 0.9188
```

```r
cat("--- Sign Test ---\n")
```

```
## --- Sign Test ---
```

```r
cat("H0: P(above 0.5) = 0.5\n")
```

```
## H0: P(above 0.5) = 0.5
```

```r
cat("H1: P(above 0.5) != 0.5\n\n")
```

```
## H1: P(above 0.5) != 0.5
```

```r
n_pos  <- sum(diffs > 0)
n_neg  <- sum(diffs < 0)
n_ties <- sum(diffs == 0)
n_eff  <- n_pos + n_neg
cat(sprintf("Positives (+): %d  |  Negatives (-): %d  |  Ties: %d\n",
            n_pos, n_neg, n_ties))
```

```
## Positives (+): 4  |  Negatives (-): 6  |  Ties: 1
```

```r
sign_p <- binom.test(n_pos, n_eff, p = 0.5)$p.value
cat(sprintf("Sign Test p = %.4f\n\n", sign_p))
```

```
## Sign Test p = 0.7539
```

```r
cat("INTERPRETATION:\n")
```

```
## INTERPRETATION:
```

```r
cat(sprintf("- Wilcoxon: V = %.0f, p = %.4f => Fail to reject H0.\n",
            wsr$statistic, wsr$p.value))
```

```
## - Wilcoxon: V = 26, p = 0.9188 => Fail to reject H0.
```

```r
cat(sprintf("- Sign Test: %d+, %d-, p = %.4f => Fail to reject H0.\n",
            n_pos, n_neg, sign_p))
```

```
## - Sign Test: 4+, 6-, p = 0.7539 => Fail to reject H0.
```

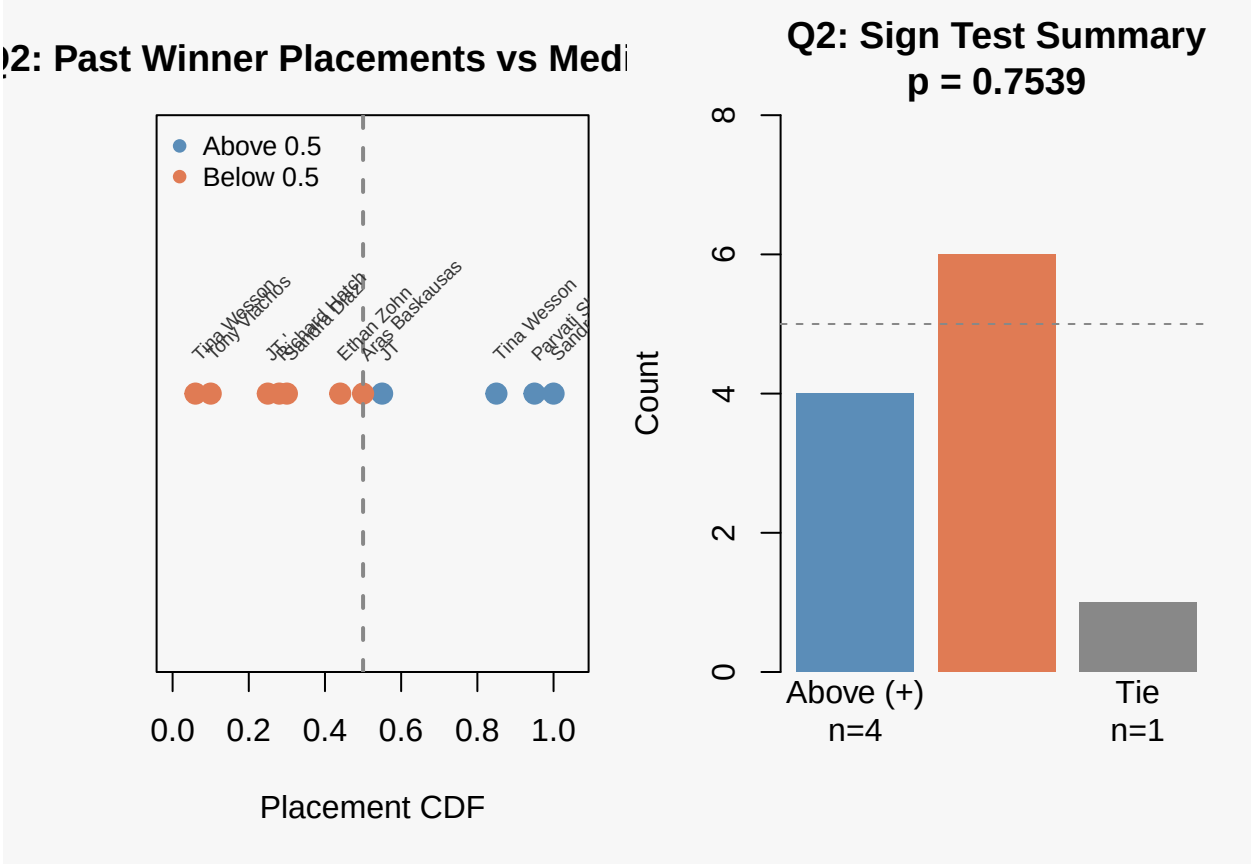
```r
cat("- Past winners do NOT perform significantly differently\n")
```

```
## - Past winners do NOT perform significantly differently
```

```r
cat("  from average when returning to play again.\n")
```

```
##   from average when returning to play again.
```

```r
# --- Q2 PLOT ---
par(mfrow = c(1, 2), bg = col_bg, mar = c(5, 4, 3, 1))
dot_cols <- ifelse(q2$Placement_CDF > 0.5, col_new, col_ret)
plot(q2$Placement_CDF, rep(1, nrow(q2)),
     col = dot_cols, pch = 19, cex = 1.4,
     xlim = c(0, 1.05), ylim = c(0.5, 1.5),
     xlab = "Placement CDF", yaxt = "n", ylab = "",
     main = "Q2: Past Winner Placements vs Median (0.5)")
abline(v = 0.5, col = col_mid, lty = 2, lwd = 2)
text(q2$Placement_CDF, rep(1, nrow(q2)) + 0.07,
     q2$Player, cex = 0.55, srt = 45, adj = 0, col = "#333333")
legend("topleft", legend = c("Above 0.5", "Below 0.5"),
       col = c(col_new, col_ret), pch = 19, bty = "n", cex = 0.8)
counts <- c(n_pos, n_neg, n_ties)
bar_labels <- c(paste0("Above (+)\nn=", n_pos),
                paste0("Below (-)\nn=", n_neg),
                paste0("Tie\nn=", n_ties))
barplot(counts, names.arg = bar_labels,
        col = c(col_new, col_ret, col_mid), border = NA,
        ylim = c(0, max(counts) + 2), ylab = "Count",
        main = sprintf("Q2: Sign Test Summary\np = %.4f", sign_p))
abline(h = n_eff / 2, lty = 2, col = col_mid)
```




```
# ============================================================
# QUESTION 3: Do 3+ time players place differently than
#             2-time players in all-returning seasons?
#             Seasons as blocks: 20, 34, 40, 50
#             Test: Friedman Test (seasons = blocks)
# ============================================================
cat("\n\n", strrep("=", 60), "\n")
```

```
##
##
##  ============================================================
```

```
cat("QUESTION 3: Friedman Test\n")
```

```
## QUESTION 3: Friedman Test
```

```
cat("Do 3+ timers place differently than 2-timers across seasons?\n")
```

```
## Do 3+ timers place differently than 2-timers across seasons?
```

```
cat(strrep("=", 60), "\n\n")
```

```
## ============================================================
```

```r
q3 <- read.csv("Survivor_Q3.csv")

cat("--- Assumption 1: Normality (Shapiro-Wilk by group) ---\n")
```

```
## --- Assumption 1: Normality (Shapiro-Wilk by group) ---
```

```r
for (g in c(0, 1)) {
  sub   <- q3[q3$Three. == g, "CDF"]
  label <- ifelse(g == 1, "3+ timers", "2-timers ")
  sw    <- shapiro.test(sub)
  cat(sprintf("%s  W = %.4f,  p = %.4f\n", label, sw$statistic, sw$p.value))
}
```

```
## 2-timers   W = 0.9452,  p = 0.0200
## 3+ timers  W = 0.9381,  p = 0.0600
```

```r
cat("\n--- Assumption 2: Homogeneity of Variance (Bartlett's Test) ---\n")
```

```
##
## --- Assumption 2: Homogeneity of Variance (Bartlett's Test) ---
```

```r
btest <- bartlett.test(CDF ~ Three., data = q3)
cat(sprintf("Bartlett's test: K2 = %.4f,  p = %.4f\n\n",
            btest$statistic, btest$p.value))
```

```
## Bartlett's test: K2 = 0.0909,  p = 0.7630
```

```r
cat("CONCLUSION: Data are not normally distributed and/or variances\n")
```

```
## CONCLUSION: Data are not normally distributed and/or variances
```

```r
cat("are unequal. With seasons as natural blocks, the Friedman test\n")
```

```
## are unequal. With seasons as natural blocks, the Friedman test
```

```r
cat("is the appropriate nonparametric blocked design test.\n\n")
```

```
## is the appropriate nonparametric blocked design test.
```

```r
seasons_q3 <- c(20, 34, 40, 50)
medians <- matrix(NA, nrow = length(seasons_q3), ncol = 2,
                  dimnames = list(paste0("S", seasons_q3),
                                  c("2-timer", "3+-timer")))
for (i in seq_along(seasons_q3)) {
  s   <- seasons_q3[i]
  sub <- q3[q3$Season == s, ]
  medians[i, 1] <- median(sub[sub$Three. == 0, "CDF"])
  medians[i, 2] <- median(sub[sub$Three. == 1, "CDF"])
}
cat("--- Median CDF per Group per Season ---\n")
```

```
## --- Median CDF per Group per Season ---
```

```
print(round(medians, 4))
```

```
##     2-timer 3+-timer
## S20  0.5000    0.600
## S34  0.6500    0.325
## S40  0.7000    0.300
## S50  0.4167    0.625
```

```
cat("\n--- Friedman Test ---\n")
```

```
##
## --- Friedman Test ---
```

```
cat("H0: No difference in placement between 2-timers and 3+ timers\n")
```

```
## H0: No difference in placement between 2-timers and 3+ timers
```

```
cat("H1: There IS a difference\n\n")
```

```
## H1: There IS a difference
```

```
friedman_result <- friedman.test(medians)
cat(sprintf("Chi2 = %.4f,  df = %d,  p = %.4f\n\n",
            friedman_result$statistic,
            friedman_result$parameter,
            friedman_result$p.value))
```

```
## Chi2 = 0.0000,  df = 1,  p = 1.0000
```

```
cat("--- Supporting: Wilcoxon Rank-Sum per Season ---\n")
```

```
## --- Supporting: Wilcoxon Rank-Sum per Season ---
```

```
cat(sprintf("%-8s %-5s %-5s %-8s %-8s\n", "Season","n(2)","n(3+)","W","p-value"))
```

```
## Season   n(2)  n(3+) W        p-value
```

```
cat(strrep("-", 38), "\n")
```

```
## --------------------------------------
```

```r
for (s in seasons_q3) {
  sub  <- q3[q3$Season == s, ]
  grp0 <- sub[sub$Three. == 0, "CDF"]
  grp1 <- sub[sub$Three. == 1, "CDF"]
  if (length(grp0) > 0 & length(grp1) > 0) {
    wt <- wilcox.test(grp0, grp1, exact = FALSE)
    cat(sprintf("S%-7s %-5d %-5d %-8.3f %-8.4f\n",
                s, length(grp0), length(grp1), wt$statistic, wt$p.value))
  }
}
```

```
## S20      11    9     44.000   0.7040
## S34      12    8     74.000   0.0491
## S40      11    9     66.000   0.2240
## S50      17    7     48.500   0.4929
```

```r
cat("\nINTERPRETATION:\n")
```

```
##
## INTERPRETATION:
```

```r
cat("- Friedman test treats each season as a block.\n")
```

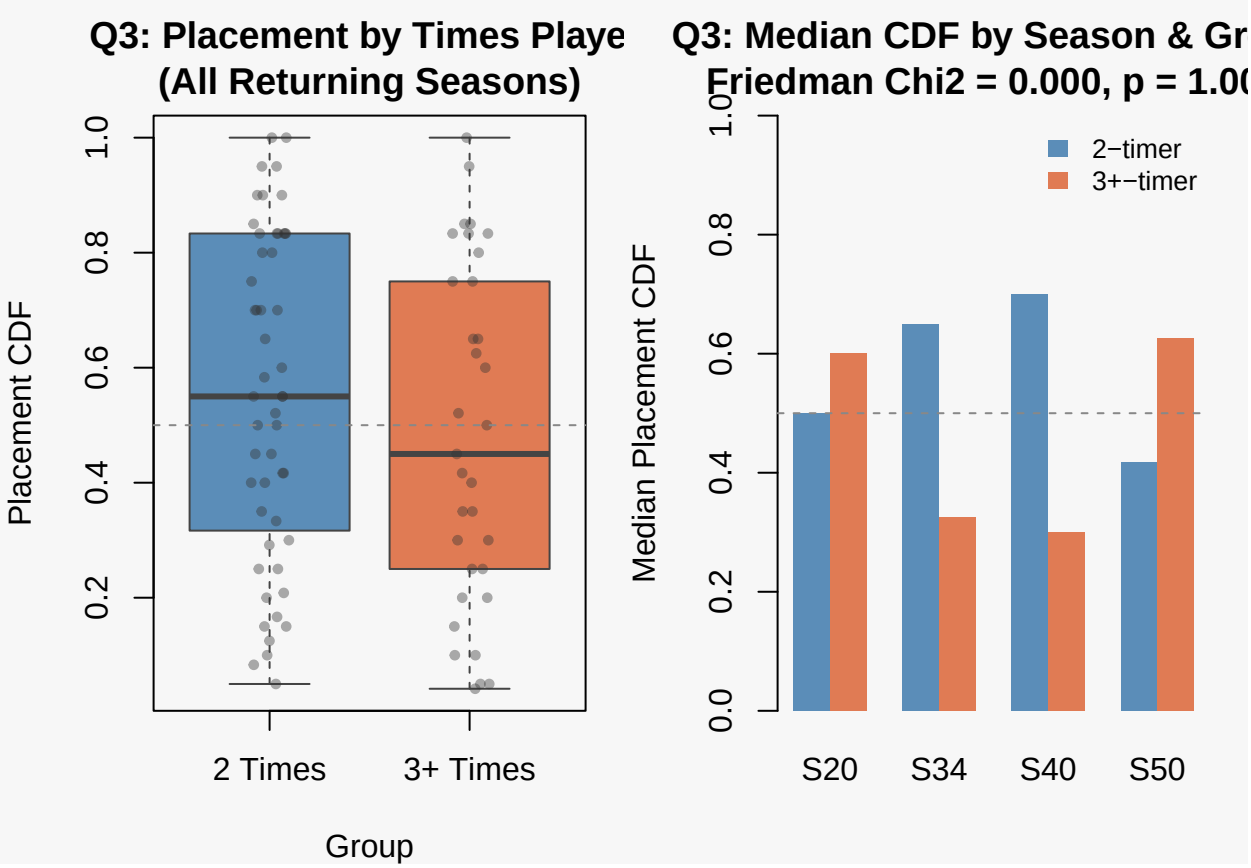
```
## - Friedman test treats each season as a block.
```

```r
if (friedman_result$p.value < 0.05) {
  cat("- p < 0.05: Significant difference between 3+ timers and 2-timers.\n")
} else {
  cat("- p > 0.05: No significant difference between groups across blocks.\n")
  cat("  3+ time players do not place significantly differently\n")
  cat("  than 2-time players in all-returning seasons.\n")
}
```

```
## - p > 0.05: No significant difference between groups across blocks.
##   3+ time players do not place significantly differently
##   than 2-time players in all-returning seasons.
```

```r
# --- Q3 PLOT ---
par(mfrow = c(1, 2), bg = col_bg, mar = c(4, 4, 3, 1))
q3$Group <- ifelse(q3$Three. == 1, "3+ Times", "2 Times")
boxplot(CDF ~ Group, data = q3,
        col = c(col_new, col_ret), border = "#444444",
        ylab = "Placement CDF",
        main = "Q3: Placement by Times Played\n(All Returning Seasons)")
stripchart(CDF ~ Group, data = q3, method = "jitter",
           pch = 19, cex = 0.6, col = "#33333366", vertical = TRUE, add = TRUE)
abline(h = 0.5, lty = 2, col = col_mid)
barplot(t(medians), beside = TRUE,
        col = c(col_new, col_ret), border = NA,
        ylim = c(0, 1), ylab = "Median Placement CDF",
        main = sprintf("Q3: Median CDF by Season & Group\nFriedman Chi2 = %.3f, p = %.3f",
```

```
                      friedman_result$statistic, friedman_result$p.value))
legend("topright", legend = c("2-timer", "3+-timer"),
       fill = c(col_new, col_ret), border = NA, bty = "n", cex = 0.85)
abline(h = 0.5, lty = 2, col = col_mid)
```




```
# ============================================================
# QUESTION 4: Do returning players place better than
#             non-returning players?
#             Block 1 (Half Returnee): 16, 26, 27
#             Block 2 (Captain): 11, 22, 25, 38
#             Test: Spearman Rank Correlation
# ============================================================
cat("\n\n", strrep("=", 60), "\n")
```

```
##
##
##  ============================================================
```

```
cat("QUESTION 4: Spearman Rank Correlation\n")
```

```
## QUESTION 4: Spearman Rank Correlation
```

```r
cat("Do returning players place better than new players?\n")
```

```
## Do returning players place better than new players?
```

```r
cat("Block 1 - Half Returnee Seasons: 16, 26, 27\n")
```

```
## Block 1 - Half Returnee Seasons: 16, 26, 27
```

```r
cat("Block 2 - Captain Seasons: 11, 22, 25, 38\n")
```

```
## Block 2 - Captain Seasons: 11, 22, 25, 38
```

```r
cat(strrep("=", 60), "\n\n")
```

```
## ============================================================
```

```r
q4 <- read.csv("Survivor_Q4.csv")

cat("--- Assumption 1: Normality (Shapiro-Wilk by Block) ---\n")
```

```
## --- Assumption 1: Normality (Shapiro-Wilk by Block) ---
```

```r
for (b in c(1, 2)) {
  sub   <- q4[q4$Block == b, "CDF"]
  label <- paste0("Block ", b)
  sw    <- shapiro.test(sub)
  cat(sprintf("%s: W = %.4f,  p = %.4f\n", label, sw$statistic, sw$p.value))
}
```

```
## Block 1: W = 0.9251,  p = 0.0364
## Block 2: W = 0.9012,  p = 0.1390
```

```r
cat("\n--- Assumption 2: Monotonic Relationship ---\n")
```

```
##
## --- Assumption 2: Monotonic Relationship ---
```

```r
cat("Placement CDF vs Block is a ranking relationship;\n")
```

```
## Placement CDF vs Block is a ranking relationship;
```

```r
cat("Spearman does not require linearity, only monotonicity.\n\n")
```

```
## Spearman does not require linearity, only monotonicity.
```

```r
cat("CONCLUSION: Normality violated in at least one block.\n")
```

```
## CONCLUSION: Normality violated in at least one block.
```

```r
cat("=> Spearman Rank Correlation is the appropriate test.\n\n")
```

```
## => Spearman Rank Correlation is the appropriate test.
```

```r
cat("--- Spearman Rank Correlation: CDF ~ Block ---\n")
```

```
## --- Spearman Rank Correlation: CDF ~ Block ---
```

```r
cat("H0: rho = 0 (no association between block type and placement)\n")
```

```
## H0: rho = 0 (no association between block type and placement)
```

```r
cat("H1: rho != 0\n\n")
```

```
## H1: rho != 0
```

```r
spr_q4 <- cor.test(q4$CDF, q4$Block, method = "spearman", exact = FALSE)
cat(sprintf("rho = %.4f,  S = %.1f,  p = %.4f\n\n",
            spr_q4$estimate, spr_q4$statistic, spr_q4$p.value))
```

```
## rho = -0.0102,  S = 13379.3,  p = 0.9482
```

```r
cat("--- Descriptive Statistics by Block ---\n")
```

```
## --- Descriptive Statistics by Block ---
```

```r
cat(sprintf("%-10s %-5s %-8s %-8s\n", "Block", "n", "Median", "Mean"))
```

```
## Block      n     Median   Mean
```

```r
cat(strrep("-", 33), "\n")
```

```
## ---------------------------------
```

```r
for (b in c(1, 2)) {
  sub <- q4[q4$Block == b, "CDF"]
  cat(sprintf("Block %-4d %-5d %-8.4f %-8.4f\n",
              b, length(sub), median(sub), mean(sub)))
}
```

```
## Block 1    30    0.6000   0.5817
## Block 2    13    0.5556   0.5855
```

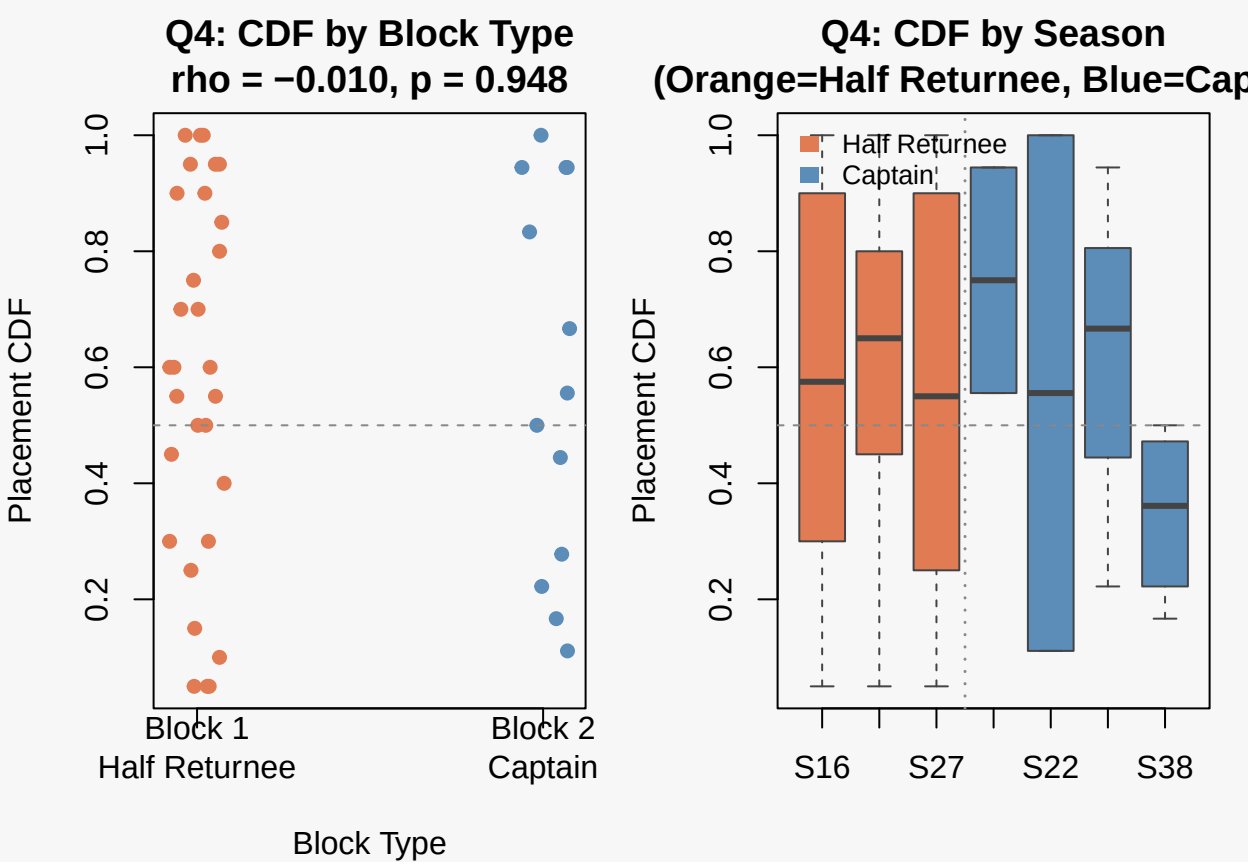
```r
cat("\nINTERPRETATION:\n")
```

```
##
## INTERPRETATION:
```

```r
if (spr_q4$p.value < 0.05) {
  cat(sprintf("- rho = %.4f, p = %.4f: Significant association.\n",
              spr_q4$estimate, spr_q4$p.value))
} else {
  cat(sprintf("- rho = %.4f, p = %.4f: No significant association.\n",
              spr_q4$estimate, spr_q4$p.value))
  cat("- Block type (Half Returnee vs Captain) is not significantly\n")
  cat("  correlated with placement CDF.\n")
}
```

```
## - rho = -0.0102, p = 0.9482: No significant association.
## - Block type (Half Returnee vs Captain) is not significantly
##   correlated with placement CDF.
```

```r
# --- Q4 PLOT ---
par(mfrow = c(1, 2), bg = col_bg, mar = c(4, 4, 3, 1))
set.seed(42)
jit <- jitter(q4$Block, amount = 0.08)
plot(jit, q4$CDF,
     col = ifelse(q4$Block == 1, col_ret, col_new),
     pch = 19, cex = 0.9, xaxt = "n",
     xlab = "Block Type", ylab = "Placement CDF",
     main = sprintf("Q4: CDF by Block Type\nrho = %.3f, p = %.3f",
                    spr_q4$estimate, spr_q4$p.value))
axis(1, at = c(1, 2), labels = c("Block 1\nHalf Returnee", "Block 2\nCaptain"))
abline(h = 0.5, lty = 2, col = col_mid)
season_order <- c(16, 26, 27, 11, 22, 25, 38)
season_cols  <- c(rep(col_ret, 3), rep(col_new, 4))
cdfs_by_season <- lapply(season_order, function(s) q4[q4$Season == s, "CDF"])
boxplot(cdfs_by_season, names = paste0("S", season_order),
        col = season_cols, border = "#444444",
        ylab = "Placement CDF",
        main = "Q4: CDF by Season\n(Orange=Half Returnee, Blue=Captain)")
abline(h = 0.5, lty = 2, col = col_mid)
abline(v = 3.5, lty = 3, col = "#888888", lwd = 1.5)
legend("topleft", legend = c("Half Returnee", "Captain"),
       fill = c(col_ret, col_new), border = NA, bty = "n", cex = 0.8)
```




```
# ============================================================
# QUESTION 5: Is there a difference between new era and old era
#             players' performance in Season 50?
#             Test: Wilcoxon Rank-Sum (Mann-Whitney U)
# ============================================================
cat("\n\n", strrep("=", 60), "\n")
```

```
##
##
##  ============================================================
```

```
cat("QUESTION 5: Wilcoxon Rank-Sum Test (Mann-Whitney U)\n")
```

```
## QUESTION 5: Wilcoxon Rank-Sum Test (Mann-Whitney U)
```

```
cat("New Era vs Old Era players in Season 50\n")
```

```
## New Era vs Old Era players in Season 50
```

```
cat(strrep("=", 60), "\n\n")
```

```
## ============================================================
```

```r
q5 <- read.csv("Survivor_Q5.csv")
new_era <- q5[q5$New.School == 1, "CDF"]
old_era <- q5[q5$New.School == 0, "CDF"]

cat("--- Assumption 1: Normality (Shapiro-Wilk by Group) ---\n")
```

```
## --- Assumption 1: Normality (Shapiro-Wilk by Group) ---
```

```r
sw_new <- shapiro.test(new_era)
sw_old <- shapiro.test(old_era)
cat(sprintf("New Era players: W = %.4f,  p = %.4f\n",
            sw_new$statistic, sw_new$p.value))
```

```
## New Era players: W = 0.8197,  p = 0.0158
```

```r
cat(sprintf("Old Era players: W = %.4f,  p = %.4f\n\n",
            sw_old$statistic, sw_old$p.value))
```

```
## Old Era players: W = 0.9023,  p = 0.1700
```

```r
cat("--- Assumption 2: Homogeneity of Variance ---\n")
```

```
## --- Assumption 2: Homogeneity of Variance ---
```

```r
btest_q5 <- bartlett.test(CDF ~ New.School, data = q5)
cat(sprintf("Bartlett's test: K2 = %.4f,  p = %.4f\n\n",
            btest_q5$statistic, btest_q5$p.value))
```

```
## Bartlett's test: K2 = 0.1949,  p = 0.6589
```

```r
cat("--- Assumption 3: Sample Size ---\n")
```

```
## --- Assumption 3: Sample Size ---
```

```r
cat(sprintf("New Era: n = %d  |  Old Era: n = %d\n\n",
            length(new_era), length(old_era)))
```

```
## New Era: n = 12  |  Old Era: n = 12
```

```r
cat("CONCLUSION: At least one group violates normality.\n")
```

```
## CONCLUSION: At least one group violates normality.
```

```r
cat("A two-sample t-test is NOT appropriate.\n")
```

```
## A two-sample t-test is NOT appropriate.
```

```r
cat("=> Wilcoxon Rank-Sum (Mann-Whitney U) is the correct test.\n\n")
```

```
## => Wilcoxon Rank-Sum (Mann-Whitney U) is the correct test.
```

```r
cat("--- Wilcoxon Rank-Sum Test ---\n")
```

```
## --- Wilcoxon Rank-Sum Test ---
```

```r
cat("H0: Distribution of placement is equal for New and Old Era players\n")
```

```
## H0: Distribution of placement is equal for New and Old Era players
```

```r
cat("H1: Distributions differ\n\n")
```

```
## H1: Distributions differ
```

```r
wrs_q5 <- wilcox.test(new_era, old_era, exact = FALSE)
cat(sprintf("W = %.1f,  p = %.4f\n\n", wrs_q5$statistic, wrs_q5$p.value))
```

```
## W = 66.0,  p = 0.7440
```

```r
cat("--- Descriptive Statistics ---\n")
```

```
## --- Descriptive Statistics ---
```

```r
cat(sprintf("New Era: Median = %.4f,  Mean = %.4f\n",
            median(new_era), mean(new_era)))
```

```
## New Era: Median = 0.4167,  Mean = 0.5000
```

```r
cat(sprintf("Old Era: Median = %.4f,  Mean = %.4f\n\n",
            median(old_era), mean(old_era)))
```

```
## Old Era: Median = 0.5521,  Mean = 0.5417
```

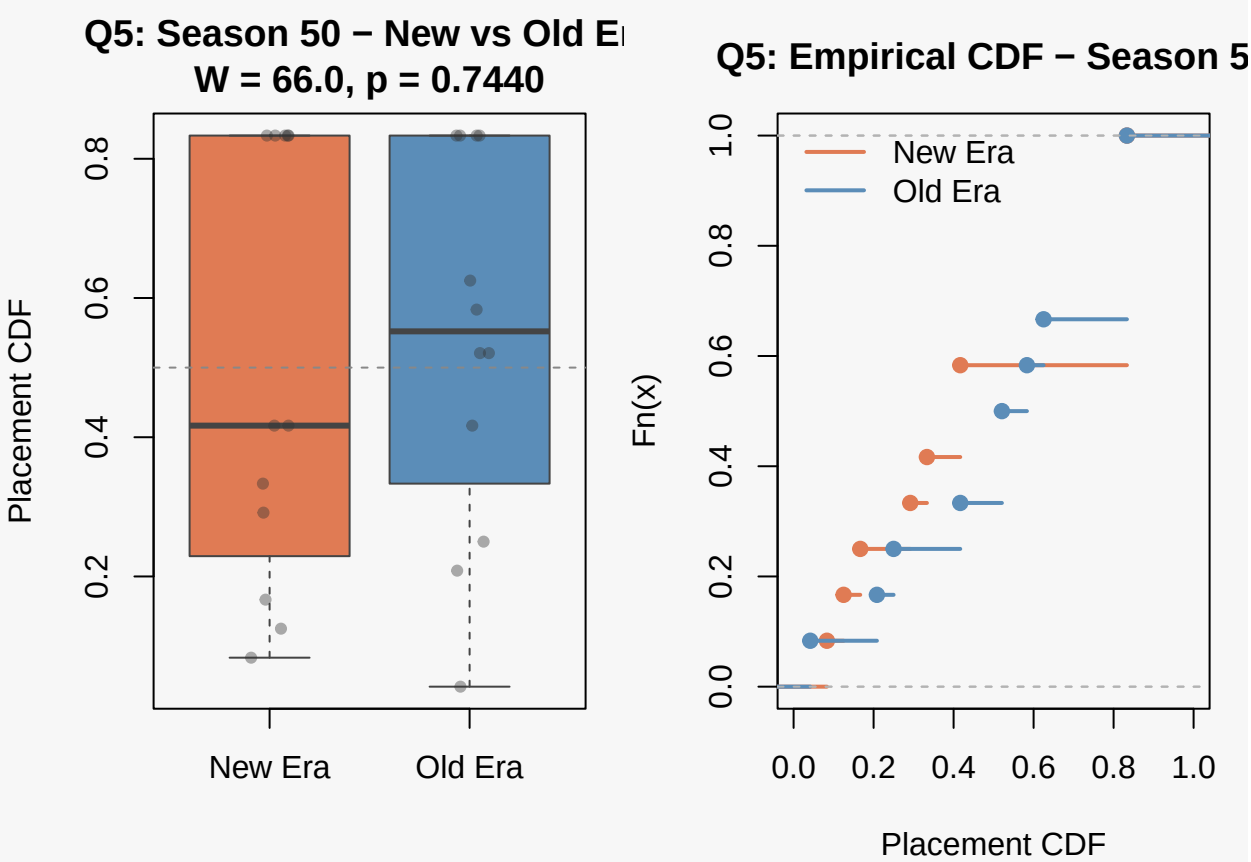
```r
cat("INTERPRETATION:\n")
```

```
## INTERPRETATION:
```

```r
if (wrs_q5$p.value < 0.05) {
  cat("- p < 0.05: Significant difference in placement between eras.\n")
} else {
  cat("- p > 0.05: No significant difference in placement between eras.\n")
  cat("- New era and old era players performed similarly in Season 50.\n")
}
```

```
## - p > 0.05: No significant difference in placement between eras.
## - New era and old era players performed similarly in Season 50.
```

```r
# --- Q5 PLOT ---
par(mfrow = c(1, 2), bg = col_bg, mar = c(4, 4, 3, 1))
boxplot(list("New Era" = new_era, "Old Era" = old_era),
        col = c(col_ret, col_new), border = "#444444",
        ylab = "Placement CDF",
        main = sprintf("Q5: Season 50 - New vs Old Era\nW = %.1f, p = %.4f",
                       wrs_q5$statistic, wrs_q5$p.value))
stripchart(list("New Era" = new_era, "Old Era" = old_era),
           method = "jitter", pch = 19, cex = 0.7,
           col = "#33333366", vertical = TRUE, add = TRUE)
abline(h = 0.5, lty = 2, col = col_mid)
plot(ecdf(new_era), col = col_ret, lwd = 2,
     main = "Q5: Empirical CDF - Season 50",
     xlab = "Placement CDF", ylab = "Fn(x)", xlim = c(0, 1))
lines(ecdf(old_era), col = col_new, lwd = 2)
legend("topleft", legend = c("New Era", "Old Era"),
       col = c(col_ret, col_new), lwd = 2, bty = "n")
```




```r
# ============================================================
# QUESTION 6: Is there a difference between Old Era and New Era
#             players in Season 40?
#             Test: Wilcoxon Rank-Sum (Mann-Whitney U)
# ============================================================
cat("\n\n", strrep("=", 60), "\n")
```

```
##
##
##  ============================================================
```

```
cat("QUESTION 6: Wilcoxon Rank-Sum Test (Mann-Whitney U)\n")
```

```
## QUESTION 6: Wilcoxon Rank-Sum Test (Mann-Whitney U)
```

```
cat("OG Era vs Modern Era players in Season 40\n")
```

```
## OG Era vs Modern Era players in Season 40
```

```
cat(strrep("=", 60), "\n\n")
```

```
## ============================================================
```

```
q6 <- read.csv("Survivor_Q6.csv")
og_era  <- q6[q6$OG.Era == 1, "CDF"]
mod_era <- q6[q6$OG.Era == 0, "CDF"]

cat("--- Assumption 1: Normality (Shapiro-Wilk by Group) ---\n")
```

```
## --- Assumption 1: Normality (Shapiro-Wilk by Group) ---
```

```
sw_og  <- shapiro.test(og_era)
sw_mod <- shapiro.test(mod_era)
cat(sprintf("OG Era players:     W = %.4f,  p = %.4f\n",
            sw_og$statistic, sw_og$p.value))
```

```
## OG Era players:     W = 0.9730,  p = 0.9206
```

```
cat(sprintf("Modern Era players: W = %.4f,  p = %.4f\n\n",
            sw_mod$statistic, sw_mod$p.value))
```

```
## Modern Era players: W = 0.9651,  p = 0.8531
```

```
cat("--- Assumption 2: Homogeneity of Variance ---\n")
```

```
## --- Assumption 2: Homogeneity of Variance ---
```

```
btest_q6 <- bartlett.test(CDF ~ OG.Era, data = q6)
cat(sprintf("Bartlett's test: K2 = %.4f,  p = %.4f\n\n",
            btest_q6$statistic, btest_q6$p.value))
```

```
## Bartlett's test: K2 = 0.5417,  p = 0.4617
```

```r
cat("--- Assumption 3: Sample Size ---\n")
```

```
## --- Assumption 3: Sample Size ---
```

```r
cat(sprintf("OG Era: n = %d  |  Modern Era: n = %d\n\n",
            length(og_era), length(mod_era)))
```

```
## OG Era: n = 8  |  Modern Era: n = 12
```

```r
cat("CONCLUSION: Normality violated and/or small n.\n")
```

```
## CONCLUSION: Normality violated and/or small n.
```

```r
cat("=> Wilcoxon Rank-Sum (Mann-Whitney U) is the correct test.\n\n")
```

```
## => Wilcoxon Rank-Sum (Mann-Whitney U) is the correct test.
```

```r
cat("--- Wilcoxon Rank-Sum Test ---\n")
```

```
## --- Wilcoxon Rank-Sum Test ---
```

```r
cat("H0: Distribution of placement is equal for OG and Modern Era\n")
```

```
## H0: Distribution of placement is equal for OG and Modern Era
```

```r
cat("H1: Distributions differ\n\n")
```

```
## H1: Distributions differ
```

```r
wrs_q6 <- wilcox.test(og_era, mod_era, exact = FALSE)
cat(sprintf("W = %.1f,  p = %.4f\n\n", wrs_q6$statistic, wrs_q6$p.value))
```

```
## W = 2.0,  p = 0.0004
```

```r
cat("--- Descriptive Statistics ---\n")
```

```
## --- Descriptive Statistics ---
```

```r
cat(sprintf("OG Era:     Median = %.4f,  Mean = %.4f\n",
            median(og_era), mean(og_era)))
```

```
## OG Era:     Median = 0.2250,  Mean = 0.2375
```

```r
cat(sprintf("Modern Era: Median = %.4f,  Mean = %.4f\n\n",
            median(mod_era), mean(mod_era)))
```

```
## Modern Era: Median = 0.7250,  Mean = 0.7250
```

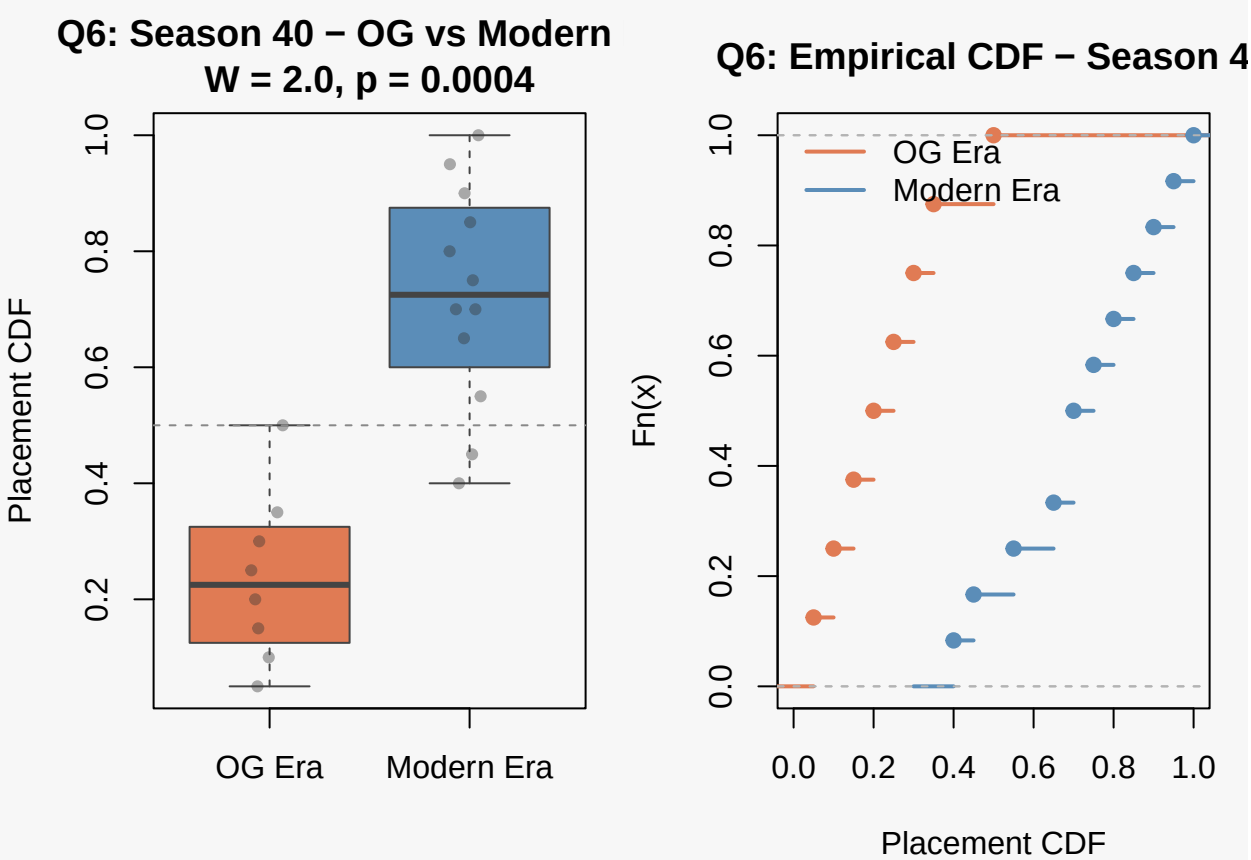
```r
cat("INTERPRETATION:\n")
```

```
## INTERPRETATION:
```

```r
if (wrs_q6$p.value < 0.05) {
  cat("- p < 0.05: Significant difference in placement between eras.\n")
} else {
  cat("- p > 0.05: No significant difference in placement between eras.\n")
  cat("- OG Era and Modern Era players performed similarly in Season 40.\n")
}
```

```
## - p < 0.05: Significant difference in placement between eras.
```

```r
# --- Q6 PLOT ---
par(mfrow = c(1, 2), bg = col_bg, mar = c(4, 4, 3, 1))
boxplot(list("OG Era" = og_era, "Modern Era" = mod_era),
        col = c(col_ret, col_new), border = "#444444",
        ylab = "Placement CDF",
        main = sprintf("Q6: Season 40 - OG vs Modern Era\nW = %.1f, p = %.4f",
                       wrs_q6$statistic, wrs_q6$p.value))
stripchart(list("OG Era" = og_era, "Modern Era" = mod_era),
           method = "jitter", pch = 19, cex = 0.7,
           col = "#33333366", vertical = TRUE, add = TRUE)
abline(h = 0.5, lty = 2, col = col_mid)
plot(ecdf(og_era), col = col_ret, lwd = 2,
     main = "Q6: Empirical CDF - Season 40",
     xlab = "Placement CDF", ylab = "Fn(x)", xlim = c(0, 1))
lines(ecdf(mod_era), col = col_new, lwd = 2)
legend("topleft", legend = c("OG Era", "Modern Era"),
       col = c(col_ret, col_new), lwd = 2, bty = "n")
```




```
# ============================================================
# QUESTION 7: Do returnees outperform new players across ALL
#             mixed seasons (half-returnee + captain)?
#             Data: Q4 all seasons (Blocks 1 & 2)
#             New player avg CDFs provided per season:
#               S11: n=16, avg=0.50  (Captain)
#               S16: n=16, avg=0.52  (Half Returnee) -- est.
#               S22: n=16, avg=0.52  (Captain)
#               S23: n=16, avg=0.48  (Captain)
#               S25: n=15, avg=0.51  (Captain)
#               S26: n=16, avg=0.52  (Half Returnee) -- est.
#               S27: n=16, avg=0.52  (Half Returnee) -- est.
#               S38: n=14, avg=0.58  (Captain)
#             Test: Wilcoxon Signed-Rank (paired by season:
#                   returnee mean vs new player mean as matched pairs)
# ============================================================
cat("\n\n", strrep("=", 60), "\n")
```

```
##
##
##  ============================================================
```

```
cat("QUESTION 7: Wilcoxon Signed-Rank Test (Paired by Season)\n")
```

```
## QUESTION 7: Wilcoxon Signed-Rank Test (Paired by Season)
```

```r
cat("Do returnees outperform new players across mixed seasons?\n")
```

```
## Do returnees outperform new players across mixed seasons?
```

```r
cat("Seasons: 11, 16, 22, 23, 25, 26, 27, 38\n")
```

```
## Seasons: 11, 16, 22, 23, 25, 26, 27, 38
```

```r
cat(strrep("=", 60), "\n\n")
```

```
## ============================================================
```

```r
# ---- Build per-season summary table ----
# New player data (provided externally)
new_info <- data.frame(
  Season   = c(11,   16,   22,   23,   25,   26,   27,   38),
  Block    = c(2,    1,    2,    2,    2,    1,    1,    2),
  Label    = c("S11 Captain", "S16 FvF",    "S22 RI",
               "S23 SP",      "S25 PvP",    "S26 Caramoan",
               "S27 BvW",     "S38 EoE"),
  n_new    = c(16,   16,   16,   16,   15,   16,   16,   14),
  mean_new = c(0.50, 0.52, 0.52, 0.48, 0.51, 0.52, 0.52, 0.58)
)

# Returnee mean CDFs from Q4 data
ret_means <- sapply(new_info$Season, function(s) {
  sub <- q4[q4$Season == s, "CDF"]
  if (length(sub) == 0) NA else mean(sub)
})
ret_n <- sapply(new_info$Season, function(s) {
  sum(q4$Season == s)
})

new_info$n_ret    <- ret_n
new_info$mean_ret <- ret_means
new_info$diff     <- new_info$mean_ret - new_info$mean_new  # positive = returnees did better

cat("--- Season-Level Summary (Returnee Mean vs New Player Mean) ---\n")
```

```
## --- Season-Level Summary (Returnee Mean vs New Player Mean) ---
```

```r
cat(sprintf("%-14s %-6s %-6s %-8s %-6s %-8s %-8s\n",
            "Season", "Blk", "n_ret", "ret_avg", "n_new", "new_avg", "diff"))
```

```
## Season         Blk    n_ret  ret_avg  n_new  new_avg  diff
```

```r
cat(strrep("-", 58), "\n")
```

```
## ----------------------------------------------------------
```

```r
for (i in seq_len(nrow(new_info))) {
  cat(sprintf("%-14s %-6d %-6d %-8.4f %-6d %-8.4f %+.4f\n",
              new_info$Label[i], new_info$Block[i],
              new_info$n_ret[i], new_info$mean_ret[i],
              new_info$n_new[i], new_info$mean_new[i],
              new_info$diff[i]))
}
```

```
## S11 Captain    2      2      0.7500   16     0.5000   +0.2500
## S16 FvF        1      10     0.5700   16     0.5200   +0.0500
## S22 RI         2      2      0.5556   16     0.5200   +0.0356
## S23 SP         2      2      0.8889   16     0.4800   +0.4089
## S25 PvP        2      3      0.6111   15     0.5100   +0.1011
## S26 Caramoan   1      10     0.6150   16     0.5200   +0.0950
## S27 BvW        1      10     0.5600   16     0.5200   +0.0400
## S38 EoE        2      4      0.3472   14     0.5800   -0.2328
```

```r
cat(sprintf("\nOverall returnee grand mean:   %.4f\n", mean(new_info$mean_ret, na.rm=TRUE)))
```

```
##
## Overall returnee grand mean:   0.6122
```

```r
cat(sprintf("Overall new player grand mean: %.4f\n\n", mean(new_info$mean_new)))
```

```
## Overall new player grand mean: 0.5188
```

```r
# ---- Assumption Checks ----
cat("--- Assumption 1: Normality of Paired Differences (Shapiro-Wilk) ---\n")
```

```
## --- Assumption 1: Normality of Paired Differences (Shapiro-Wilk) ---
```

```r
sw_diff <- shapiro.test(new_info$diff)
cat(sprintf("Differences: W = %.4f,  p = %.4f\n\n",
            sw_diff$statistic, sw_diff$p.value))
```

```
## Differences: W = 0.9232,  p = 0.4561
```

```r
if (sw_diff$p.value < 0.05) {
  cat("CONCLUSION: Differences are NOT normally distributed (p < 0.05).\n")
  cat("A paired t-test is NOT appropriate.\n")
  cat("=> Wilcoxon Signed-Rank Test (paired) is required.\n\n")
} else {
  cat("NOTE: n = 8 seasons gives Shapiro-Wilk low power.\n")
  cat("CDF data is bounded [0,1] and ordinal in nature.\n")
  cat("=> Wilcoxon Signed-Rank is more appropriate than a paired t-test.\n\n")
}
```

```
## NOTE: n = 8 seasons gives Shapiro-Wilk low power.
## CDF data is bounded [0,1] and ordinal in nature.
## => Wilcoxon Signed-Rank is more appropriate than a paired t-test.
```

```r
cat("--- Assumption 2: Paired Design ---\n")
```

```
## --- Assumption 2: Paired Design ---
```

```r
cat("Each season provides one returnee mean and one new player mean,\n")
```

```
## Each season provides one returnee mean and one new player mean,
```

```r
cat("creating 8 natural matched pairs. The within-season pairing\n")
```

```
## creating 8 natural matched pairs. The within-season pairing
```

```r
cat("controls for season-specific difficulty, twist, and cast size.\n\n")
```

```
## controls for season-specific difficulty, twist, and cast size.
```

```r
cat("--- Assumption 3: Sample Size ---\n")
```

```
## --- Assumption 3: Sample Size ---
```

```r
cat(sprintf("n = %d pairs (one per season)\n\n", nrow(new_info)))
```

```
## n = 8 pairs (one per season)
```

```r
# ---- Wilcoxon Signed-Rank (paired: ret mean vs new mean) ----
mu0_q7 <- 0.525   # overall expected average CDF

cat("--- Wilcoxon Signed-Rank Test (Paired by Season) ---\n")
```

```
## --- Wilcoxon Signed-Rank Test (Paired by Season) ---
```

```r
cat("H0: Median difference (returnee - new) = 0  (no advantage)\n")
```

```
## H0: Median difference (returnee - new) = 0  (no advantage)
```

```r
cat("H1: Median difference > 0  (returnees outperform; one-tailed)\n\n")
```

```
## H1: Median difference > 0  (returnees outperform; one-tailed)
```

```r
wsr_q7   <- wilcox.test(new_info$mean_ret, new_info$mean_new,
                        paired = TRUE, alternative = "greater", exact = FALSE)
wsr_q7_2 <- wilcox.test(new_info$mean_ret, new_info$mean_new,
                        paired = TRUE, exact = FALSE)
cat(sprintf("V = %.1f,  p = %.4f  (one-tailed, greater)\n",
            wsr_q7$statistic, wsr_q7$p.value))
```

```
## V = 30.0,  p = 0.0537  (one-tailed, greater)
```

```r
cat(sprintf("V = %.1f,  p = %.4f  (two-tailed)\n\n",
            wsr_q7_2$statistic, wsr_q7_2$p.value))
```

```
## V = 30.0,  p = 0.1073  (two-tailed)
```

```r
# Sign breakdown of differences
n_pos_d  <- sum(new_info$diff > 0)
n_neg_d  <- sum(new_info$diff < 0)
n_tie_d  <- sum(new_info$diff == 0)
cat(sprintf("Direction of differences: %d seasons returnees > new,  %d seasons new > returnees,  %d ties
            n_pos_d, n_neg_d, n_tie_d))
```

```
## Direction of differences: 7 seasons returnees > new,  1 seasons new > returnees,  0 ties
```

```r
cat("--- Descriptive Statistics ---\n")
```

```
## --- Descriptive Statistics ---
```

```r
cat(sprintf("Mean difference (ret - new): %+.4f\n", mean(new_info$diff)))
```

```
## Mean difference (ret - new): +0.0935
```

```r
cat(sprintf("Median difference:           %+.4f\n\n", median(new_info$diff)))
```

```
## Median difference:           +0.0725
```

```r
cat("--- By Block ---\n")
```

```
## --- By Block ---
```

```r
for (b in c(1, 2)) {
  sub <- new_info[new_info$Block == b, ]
  label <- ifelse(b == 1, "Half Returnee (16, 26, 27)", "Captain (11, 22, 23, 25, 38)")
  cat(sprintf("Block %d - %s\n", b, label))
  cat(sprintf("  Mean returnee CDF: %.4f  |  Mean new player CDF: %.4f  |  Mean diff: %+.4f\n\n",
              mean(sub$mean_ret), mean(sub$mean_new), mean(sub$diff)))
}
```

```
## Block 1 - Half Returnee (16, 26, 27)
##   Mean returnee CDF: 0.5817  |  Mean new player CDF: 0.5200  |  Mean diff: +0.0617
##
## Block 2 - Captain (11, 22, 23, 25, 38)
##   Mean returnee CDF: 0.6306  |  Mean new player CDF: 0.5180  |  Mean diff: +0.1126
```

```r
cat("INTERPRETATION:\n")
```

```
## INTERPRETATION:
```

```r
if (wsr_q7$p.value < 0.05) {
  cat(sprintf("- One-tailed p = %.4f < 0.05: Reject H0.\n", wsr_q7$p.value))
  cat("- Returnees significantly outperform new players within their\n")
  cat("  shared seasons when compared as paired season-level averages.\n")
} else {
  cat(sprintf("- One-tailed p = %.4f >= 0.05: Fail to reject H0.\n", wsr_q7$p.value))
  cat("- Returnees do NOT significantly outperform new players.\n")
  cat("- While returnees edge out new players in most seasons, the\n")
  cat("  difference is not statistically significant at alpha = 0.05.\n")
  cat("- Notably Season 38 (Edge of Eden) new players avg 0.58,\n")
  cat("  outperforming returnees - pulling against the hypothesis.\n")
}
```

```
## - One-tailed p = 0.0537 >= 0.05: Fail to reject H0.
## - Returnees do NOT significantly outperform new players.
## - While returnees edge out new players in most seasons, the
##   difference is not statistically significant at alpha = 0.05.
## - Notably Season 38 (Edge of Eden) new players avg 0.58,
##   outperforming returnees - pulling against the hypothesis.
```

```r
# --- Q7 PLOT ---
par(mfrow = c(1, 2), bg = col_bg, mar = c(5, 4, 3, 1))

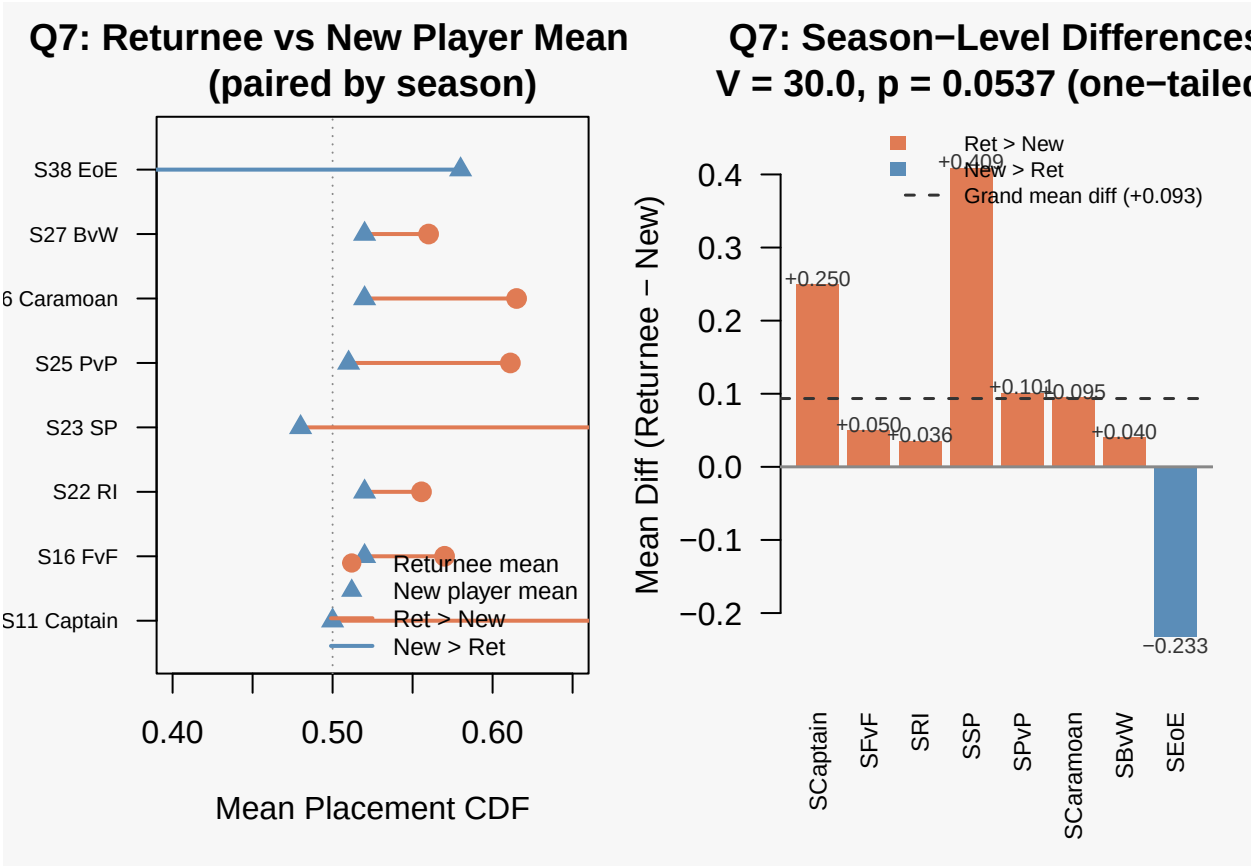
# Panel 1: paired dot + line plot per season
n_s    <- nrow(new_info)
y_pos  <- seq_len(n_s)
x_lim  <- c(0.40, 0.65)

plot(NULL, xlim = x_lim, ylim = c(0.5, n_s + 0.5),
     xlab = "Mean Placement CDF", yaxt = "n", ylab = "",
     main = "Q7: Returnee vs New Player Mean CDF\n(paired by season)")
axis(2, at = y_pos, labels = new_info$Label, las = 1, cex.axis = 0.68)
abline(v = 0.5, col = col_mid, lty = 3, lwd = 1)

# draw connecting lines first
for (i in y_pos) {
  segments(new_info$mean_new[i], i, new_info$mean_ret[i], i,
           col = ifelse(new_info$diff[i] > 0, col_ret, col_new),
           lwd = 2)
}
# draw points on top
points(new_info$mean_ret, y_pos, pch = 19, cex = 1.3, col = col_ret)
points(new_info$mean_new, y_pos, pch = 17, cex = 1.2, col = col_new)

legend("bottomright",
       legend = c("Returnee mean", "New player mean",
                  "Ret > New", "New > Ret"),
       col    = c(col_ret, col_new, col_ret, col_new),
       pch    = c(19, 17, NA, NA),
       lty    = c(NA, NA, 1, 1),
       lwd    = c(NA, NA, 2, 2),
       pt.cex = c(1.2, 1.1, NA, NA),
       bty = "n", cex = 0.72)
```

```r
# Panel 2: bar chart of differences per season
diff_cols <- ifelse(new_info$diff > 0, col_ret, col_new)
bp7 <- barplot(new_info$diff,
               names.arg = sub(".* ", "S", new_info$Label),
               col = diff_cols, border = NA,
               ylim = c(min(new_info$diff) - 0.05, max(new_info$diff) + 0.07),
               ylab = "Mean Diff (Returnee - New)",
               main = sprintf("Q7: Season-Level Differences\nV = %.1f, p = %.4f (one-tailed)",
                              wsr_q7$statistic, wsr_q7$p.value),
               las = 2, cex.names = 0.75)
abline(h = 0, col = col_mid, lwd = 1.5)
abline(h = mean(new_info$diff), col = "#333333", lty = 2, lwd = 1.5)
text(bp7, new_info$diff + ifelse(new_info$diff >= 0, 0.008, -0.012),
     sprintf("%+.3f", new_info$diff), cex = 0.65, col = "#333333")
legend("topright",
       legend = c("Ret > New", "New > Ret",
                  sprintf("Grand mean diff (%+.3f)", mean(new_info$diff))),
       fill   = c(col_ret, col_new, NA),
       border = NA,
       lty    = c(NA, NA, 2),
       lwd    = c(NA, NA, 1.5),
       col    = c(col_ret, col_new, "#333333"),
       bty = "n", cex = 0.68)
```



```r
cat("\n\n", strrep("=", 60), "\n")
```

```
##
##
##  ============================================================
```

```r
cat("ALL ANALYSES COMPLETE\n")
```

```
## ALL ANALYSES COMPLETE
```

```r
cat("Plots displayed on screen for Q1 through Q7.\n")
```

```
## Plots displayed on screen for Q1 through Q7.
```

```r
cat(strrep("=", 60), "\n")
```

```
## ============================================================
```